

Business Requirement Document (BRD)

Customer Churn Analysis & Retention Strategy

Domain: Banking Analytics | Version: 1.1

Author: Nasrul Khair (Data Analyst)

1. Project Overview

This project aims to analyze bank customer data to identify the key factors influencing customer churn and provide data-driven retention strategies. The analysis will support management in understanding customer behavior, churn patterns, and opportunities for improving engagement.

Dataset Summary

- Demographics (Age, Gender, Geography)
- Financial Indicators (Balance, Credit Score, Estimated Salary)
- Banking Behaviors (Tenure, Activity Status, Credit Card Ownership)
- Customer Status (Exited or Retained)

Attribute	Detail
Source	Local Folder (Bank_Churn.csv)
Rows	10,001 rows in the source file (10,000 customer records + 1 header row)
Type	Structured Tabular Data
Destination (Raw layer)	Snowflake Data Warehouse – Schema: RAW.BANK_CHURN

2. Business Objectives

- 1. Identify characteristics of customers likely to churn.
- 2. Quantify churn rate and segment customers by risk level.
- 3. Develop actionable insights for retention strategy.
- 4. Enable management to monitor churn KPIs through Power BI dashboards.
- 5. Establish a repeatable data pipeline from ingestion to reporting.

3. Business Questions

(To be refined during Sprint 2 – EDA)

- 1. What is the overall churn rate across all customers?
- 2. How do age, tenure, and balance influence churn likelihood?
- 3. Are inactive customers or those without credit cards more likely to churn?
- 4. Which geography or gender shows higher churn trends?
- 5. What common patterns exist among high-balance but low-activity customers?
- 6. How do credit score segments correlate with churn?
- 7. What is the retention trend by tenure segment?

4. Data Gathering

Please use the following data related to Bank Customer and Associated details:

- Source Name: Bank_Churn.csv
- Description: Customer churn dataset including demographics, credit, and account info
- Location: Local folder
- NumOfRows: 10,001 (10,000 data rows + header)

Column	Description	Example	Type	Business Use
RowNumber	Record Index	1	Integer	Ignored in analysis
CustomerId	Unique customer ID	15634602	Integer	Identifier for customer
CreditScore	Credit rating	619	Numeric	Measure of creditworthiness
GeographyId	1=France, 2=Spain, 3=Germany	1	Categorical	Region segmentation
GenderId	1=Male, 2=Female	2	Categorical	Gender-based insights
Age	Customer age	42	Numeric	Key predictor for churn
Tenure	Years with bank	7	Numeric	Measures customer loyalty
Balance	Account balance	0	Numeric	Financial indicator
NumOfProducts	Products owned	1	Numeric	Measures engagement
HasCrCard	1=Yes, 0=No	1	Binary	Product ownership
IsActiveMember	1=Yes, 0=No	1	Binary	Engagement level
EstimatedSalary	Estimated annual income	101348.88	Numeric	Income category
Exited	1=Yes, 0=No	1	Binary	Churn indicator
Bank DOJ	Date joined bank	23/3/2016	Date	Used for tenure validation

5. Data Dictionary

- RowNumber – corresponds to the record number and has no effect on the output.
- CustomerId – contains random values and has no effect on customer churn.
- Surname – has no impact on a customer's decision to leave the bank.
- CreditScore – can affect churn; a customer with a higher credit score is generally less likely to leave.

Credit Score Bands

Band	Range
Excellent	800 – 850
Very Good	740 – 799
Good	670 – 739
Fair	580 – 669
Poor	300 – 579

- Geography – a customer's location can affect their decision to leave the bank.
- Gender – worth exploring whether gender plays a role in churn.
- Age – older customers are generally less likely to leave than younger ones.
- Tenure – number of years as a client; longer-tenured clients tend to be more loyal.
- Balance – customers with higher balances tend to churn less.
- NumOfProducts – number of products purchased through the bank.
- HasCrCard – whether the customer has a credit card (1 = holder, 0 = non-holder); cardholders tend to churn less.
- IsActiveMember – active customers (1) are less likely to churn than inactive ones (0).
- EstimatedSalary – similar to balance, higher earners tend to churn less.
- Exited – whether the customer left the bank (1 = Exit, 0 = Retain).
- Bank DOJ – date the customer joined the bank.

6. Business Rules

Rule	Description
Churn Definition	Customers with Exited = 1 are churned.
Activity	IsActiveMember = 0 → Low engagement risk group.
Location Mapping	1 = France, 2 = Spain, 3 = Germany
Gender Mapping	1 = Male, 2 = Female
Data Validation	Exclude rows with missing or invalid CustomerId.
Date Handling	Convert Bank DOJ to datetime and validate Tenure.

7. Deliverables

Deliverable	Description	Owner	Related Epic
BRD	BRD & FRD	BA	Epic 1
Snowflake Database Setup	Create BANK_CHURN DB and schemas (RAW, CLEAN, ANALYTICS)	Data Engineer	Epic 2
Data Cleaning & Transformation Scripts	Python/SQL scripts for preprocessing	Data Engineer	Epic 2
Star Schema Data Model	Fact_CustomerChurn + dimension tables	Data Modeler	Epic 3
EDA Report	Insights and trend analysis	Data Analyst	Epic 3
Power BI Dashboard	Interactive churn dashboard with filters	Power BI Developer	Epic 4
Row-Level Security Setup	Restrict views by geography	BI Admin	Epic 5
Data Refresh Pipeline	Scheduled refresh via gateway	BI Admin	Epic 5
Documentation Package	ER diagram, data dictionary, process flow, KPIs	BA	Epic 6

8. Stakeholders & Responsibilities

Role	Responsibility	Primary Contact
Business Analyst	Gather requirements, create BRD/FRD, perform EDA	Nasrul
Data Engineer	Build Snowflake warehouse, manage data pipelines	Nasrul
Data Analyst	Conduct churn analysis, calculate KPIs	Nasrul
Data Modeller	Design analytical star schema	Nasrul
Power BI Developer	Develop dashboards, configure RLS and refresh	Nasrul
End Users	Consume dashboards, provide feedback	Nasrul

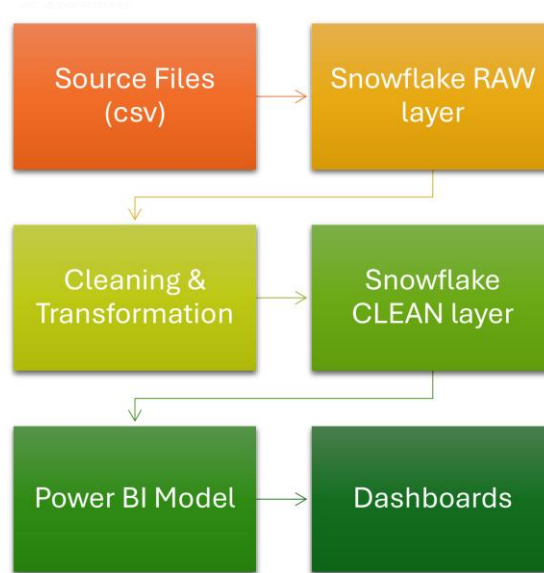
9. KPIs and Success Metrics

KPI	Description	Formula	Target / Benchmark
Customer Churn Rate	% of customers who left	$(\text{Churned} / \text{Total}) * 100$	$\leq 10\%$
Customer Retention Rate	% of customers retained	$100 - \text{Churn Rate}$	$\geq 90\%$
Active Member Ratio	% active in last 12 months	$(\text{Active} / \text{Total}) * 100$	$\geq 70\%$
Avg. Credit Score (Churned vs Retained)	Compare creditworthiness	AVG(CreditScore) by group	Retained > Churned
Avg. Balance (Churned vs Retained)	Financial profile difference	AVG(Balance) by group	Retained > Churned
High-Risk Segment %	Share predicted to churn	$(\text{Predicted_Churners} / \text{Total}) * 100$	Monitor trend ↓
Dashboard Adoption Rate	% active report users	$(\text{Active Users} / \text{Total Users}) * 100$	$\geq 80\%$
Refresh Success Rate	% successful scheduled refreshes	$(\text{Success} / \text{Total}) * 100$	100%

10. Dependencies & Constraints

Dependency	Description
Data Availability	Bank_Churn dataset must be uploaded to Snowflake before transformation.
Tool Access	Power BI Pro license and Snowflake connection credentials required.
Security	RLS must align with internal user access policy.
Environment	Stable gateway for automated refreshes.

11. Data Flow



12. Refresh Frequency

Source Name	Owner	Refresh Frequency
bank_churn.csv	Data Team	Weekly
Snowflake RAW Schema	Data Engineering	Daily
Snowflake CLEAN Schema	Data Engineering	Daily
Power BI Dataset	BI Team	Daily / On Refresh

13. KPI Mapping to Data Fields

KPI	Calculation	Related Fields
Churn Rate	(#Exited Customers / Total Customers)	Exited
Average Credit Score	AVG(CreditScore)	CreditScore
Avg. Tenure	AVG(Tenure)	Tenure
Active vs Inactive Members	COUNT(IsActiveMember)	IsActiveMember
Product Penetration	AVG(NumOfProducts)	NumOfProducts
Avg. Balance per Geography	AVG(Balance) GROUP BY GeographyID	Balance, GeographyID

14. Sprint Deliverables

- 1. Business Requirement Document (BRD)
- 2. Listed data sources, owners, refresh frequency, and access levels
- 3. Created a small data flow diagram / table mapping doc

Data Engineering & Warehouse Setup

1. Warehouse & Schemas Setup

“As a data engineer, I want to set up the data warehouse to host raw and clean data layers so that data is organized and ready.”

Goal: Build and configure Snowflake to host raw and cleaned banking data for analysis.

#	Task	Description	Output
1	Create Snowflake Database and Warehouse	Create a new database (e.g., BANK_ANALYTICS_DB) and warehouse (e.g., BANK_WH)	Database + warehouse ready
2	Create Schemas (RAW, CLEAN, ANALYTICS)	Organize data layers: raw source, cleaned/transformed, modeling layer	3 schemas created
3	Define Naming Conventions	Standards for database, schema, table, and column names (e.g., snake_case)	Naming convention document
4	Document Folder Structure	Short documentation for structure	Warehouse setup documentation

Warehouse & Database Structure

Name	Purpose	Description
BANK_WH	Project warehouse	Warehouse for banking churn analysis project
BANK_ANALYTICS_DB	Project database	Database for customer churn analysis and retention insights
BANK_ANALYTICS_DB.RAW	Raw data schema	Stores raw uploaded datasets (unmodified)
BANK_ANALYTICS_DB.CLEAN	Cleaned data schema	Stores cleaned and transformed datasets
BANK_ANALYTICS_DB.ANALYTICS	Analytics schema	Stores data models and views used by Power BI

2. Load Source Data (Raw Layer)

“As a data engineer, I want to load source data into the RAW layer in Snowflake so that it's available for cleaning and analysis.”

Validation Checks

Purpose	SQL Query
Preview First 10 records	SELECT * FROM RAW.BANK_CHURN LIMIT 10;
Validate Row Count	SELECT COUNT(*) AS total_rows FROM RAW.BANK_CHURN; -- 10,000

Results

Check	Expected	Actual	Status
Row Count	10,000 data rows	10,000 data rows (matches — file has 10,001 lines incl. header)	Passed
Column Headers	14 columns	14 columns	Passed
Data Type Consistency	Matches CSV	Matches CSV	Passed
Null Values	0 in key fields	Verification pending — see note below	Action required

3. Data Cleaning & Transformation

“As a data engineer, I want to extract, clean, and transform the data so that it's accurate, standardized, and ready for modelling.”

a. Extract RAW data from Snowflake

```
import os
import pandas as pd
from snowflake.connector import connect
import warnings
warnings.filterwarnings('ignore')

details = {
    'user': os.environ['SNOWFLAKE_USER'],
    'password': os.environ['SNOWFLAKE_PASSWORD'],
    'account': os.environ['SNOWFLAKE_ACCOUNT'],
    'warehouse': 'BANK_WH',
    'database': 'BANK_ANALYTICS_DB',
    'schema': 'RAW'
}

conn = connect(**details)
query = 'SELECT * FROM BANK_CHURN'
df = pd.read_sql(query, conn)
df.head()
```

Credentials are loaded from environment variables (or a git-ignored .env file) so they never appear in source control. If this project's history was ever pushed to a public repo with the credentials inline, rotate the password and audit the account for unexpected activity.

b. Inspect for Missing / Invalid Data

```
df.info()
df.isnull().sum()
df.describe()
```

c. Clean & Transform

```
# renaming columns
df = df.rename(columns={
    'ROWNUMBER':'row_number', 'CUSTOMERID':'customer_id',
    'CREDITSCORE':'credit_score', 'GEOGRAPHYID':'geography_id',
    'GENDERID':'gender_id', 'AGE':'age', 'TENURE':'tenure',
    'BALANCE':'balance', 'NUMOFPRODUCTS':'num_of_products',
    'HASCRCARD':'has_cr_card', 'ISACTIVEMEMBER':'is_active_member',
    'ESTIMATEDSALARY':'estimated_salary', 'EXITED':'exited',
    'BANK_DOJ':'bank_doj'
})

df['bank_doj'] = pd.to_datetime(df['bank_doj'])
df['geography_name'] = df['geography_id'].map({1:'France', 2:'Spain', 3:'Germany'})
df['gender'] = df['gender_id'].map({1:'Male', 2:'Female'})
df['retain_status'] = df['exited'].map({1:'Exit', 0:'Retain'})
df['credit_card_holder'] = df['has_cr_card'].map({1:'Credit Card Holder', 0:'Non Credit Card Holder'})
df = df.drop(columns=['row_number'])
```

d. Load Cleaned Data to Snowflake (CLEAN layer)

```
from snowflake.connector.pandas_tools import write_pandas

conn.cursor().execute("USE SCHEMA BANK_ANALYTICS_DB.CLEAN")
success, nchunks, nrows, _ = write_pandas(
    conn=conn, df=df, table_name='BANK_CHURN_CLEANED',
    schema='CLEAN', auto_create_table=True, overwrite=True
)
print(f"Upload successful: {success}, rows inserted: {nrows}")
```

Data Quality Validation

Check	Description	Expected Result	Status
Row Count	Matches RAW layer	10,000	Passed
Null Values	Cleaned	0 nulls in key columns	Passed — pending re-verification (see note above)
Category Mapping	Correct Geography/Gender	France/Spain/Germany; Male/Female	Passed
Business Logic	“Churned” correctly derived	Consistent	Passed

Data Modelling & Exploration

1. Analytical & Data Model Design

“As a data modeler, I want to design a star schema so that Power BI and analysts can efficiently query, aggregate, and visualize customer churn insights.”

Data Model Design (Star Schema)

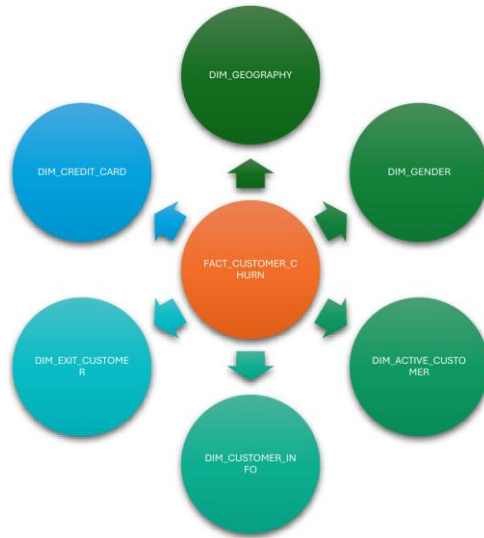


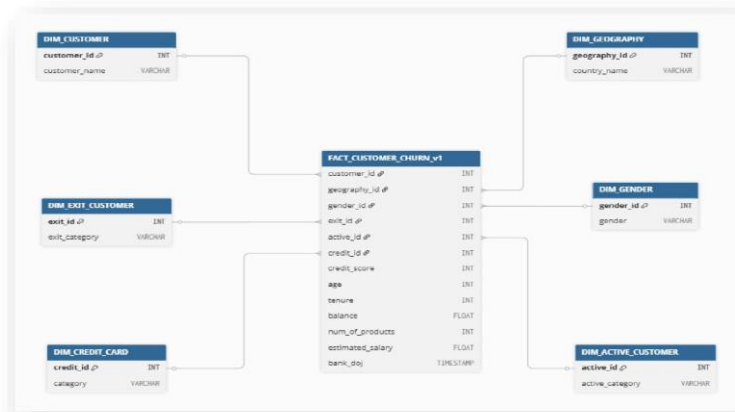
Table Definitions

FACT_CUSTOMER_CHURN_V1

Column	Type	Description
customer_id	INT	FK to DIM_CUSTOMER
geography_id	INT	FK to DIM_GEOGRAPHY
gender_id	INT	FK to DIM_GENDER
exit_id	INT	FK to DIM_EXIT_CUSTOMER
active_id	INT	FK to DIM_ACTIVE_CUSTOMER
cr_card_id	INT	FK to DIM_CREDIT_CARD
credit_score	INT	Customer credit score
age	INT	Customer age
tenure	INT	Years with bank
balance	FLOAT	Account balance
num_of_products	INT	Products owned
estimated_salary	FLOAT	Annual income
churned	VARCHAR	Exited, Retained
bank_doj	TIMESTAMP	Load timestamp

Dimension tables: DIM_CUSTOMER (customer_id PK, customer_name), DIM_GEOGRAPHY (geography_id PK, country_name), DIM_GENDER (gender_id PK, gender), DIM_EXIT_CUSTOMER (exit_id PK, exit_category), DIM_ACTIVE_CUSTOMER (active_id PK, active_category), DIM_CREDIT_CARD (credit_id PK, category).

ER Diagram



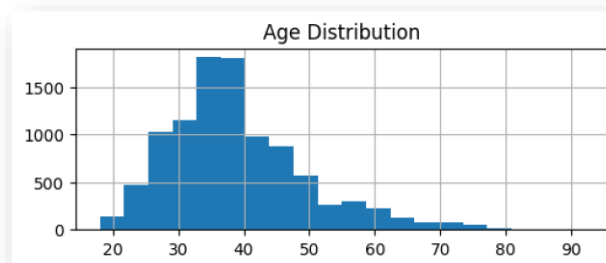
2. Exploratory Data Analysis (EDA)

“As a data analyst, I want to perform exploratory analysis on the cleaned data so that I can understand patterns, relationships, and factors contributing to customer churn.”

1. Univariate Analysis

a. Age Distribution

a) a. Age Distribution

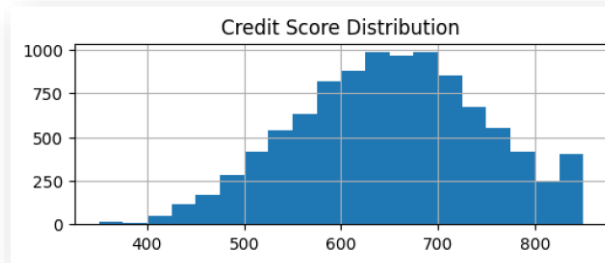


* Majority of customers are aged 35-40 years old

Majority of customers are aged 35–40 years old.

b. Credit Score Distribution

b) Credit Score Distribution

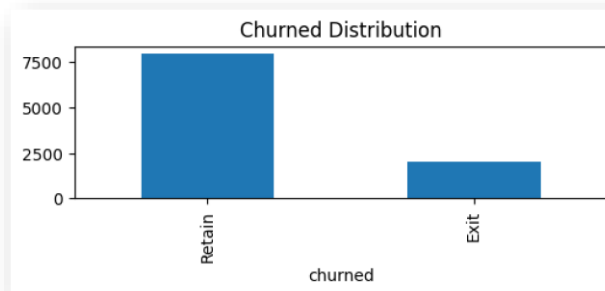


* Credit Scores cluster around 600-700 range

Credit scores cluster around the 600–700 range.

c. Churned Distribution

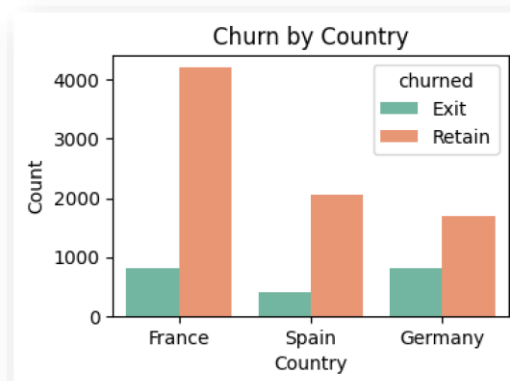
c) Churned Distribution



2. Bivariate Analysis

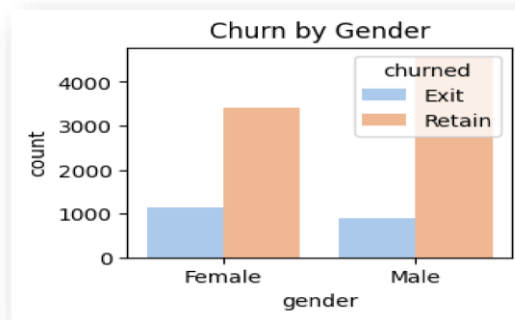
1. Churn by Geography

Churn By Geography

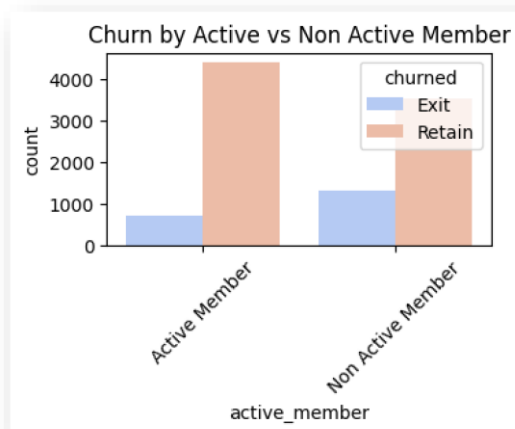


2. Churn by Gender

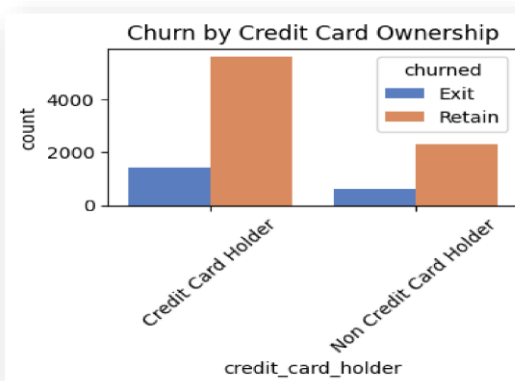
Churn By Gender



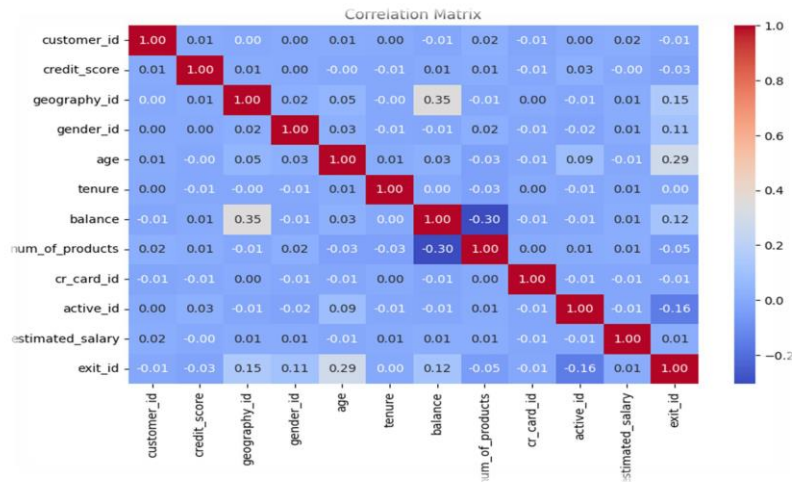
3. Churn by Active Member Status



4. Churn by Credit Card Holder Status



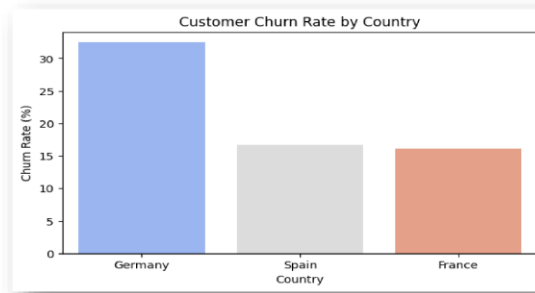
3. Correlation Matrix



Metric	Score	Interpretation
Age	0.29	Weak-to-moderate positive relationship with churn — the strongest single numeric correlation in this dataset, but still modest.
Geography	0.15	Weak; churn appears to differ by country — worth confirming via the segmentation breakdown rather than this correlation alone.
Balance	0.12	Weak; higher-balance customers show a slightly higher churn tendency — not strong enough to call a driver on its own.
Gender	0.11	Weak; treat as a hypothesis to test (e.g., chi-square), not a confirmed effect.
Active Member	-0.16	Weak-to-moderate negative relationship — directionally consistent with the bivariate chart showing active members churn less.
Estimated Salary	0.01	Negligible — essentially no linear relationship with churn.
Tenure	0.001	Negligible — no relationship with churn in this dataset.
Credit Card Ownership	-0.01	Negligible.
Credit Score	-0.03	Negligible.
Number of Products	-0.05	Negligible-to-weak.

4. Segmentation Analysis

Country	Total Customers	Churned Customers	Churn Rate %
Germany	2,509	814	32.44
Spain	2,477	413	16.67
France	5,014	810	16.15



Insights

- Germany shows the highest churn rate (32.4%), almost double that of France and Spain.
- France has the largest customer base (5,014 customers) but maintains the lowest churn rate (16.15%).
- Spain sits in between — moderate churn (16.7%) with a slightly smaller customer base than France.

3. Core Business Questions

“As a data analyst, I want to answer core business questions using SQL so that I can validate business KPIs using the cleaned data.”

Core Business Metrics Definition

KPI Name	Formula	Business Meaning
Churn Rate	$\text{SUM(Exited)} * 100 / \text{COUNT(*)}$	% of customers who left
Average Credit Score (Churned vs Retained)	AVG(CreditScore)	Credit profile by churn group
Average Tenure by Churn Group	AVG(Tenure)	How long churned customers stayed
Churn Rate by Country	Grouped churn %	Identify high-risk regions
Churn Rate by Product Holding	Grouped churn % by NumOfProducts	Cross-sell opportunity

1. Overall Churn Rate

```
SELECT
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED;
```

CHURN_RATE_PERCENTAGE
20.37

2. Churn Rate by Country

```
SELECT
  "country_name",
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL;
```

Country	Churn Rate %
France	16.15
Spain	16.67
Germany	32.44

3. Churn Rate by Gender

```

SELECT
  "gender",
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM bank_churn_cleaned
GROUP BY ALL;

```

Gender	Churn Rate %
Female	25.07
Male	16.46

4. Churn Rate by Age Group

```

SELECT
  CASE
    WHEN "age" < 30 THEN 'Under 30'
    WHEN "age" BETWEEN 30 AND 39 THEN '30s'
    WHEN "age" BETWEEN 40 AND 49 THEN '40s'
    WHEN "age" BETWEEN 50 AND 59 THEN '50s'
    ELSE '60s+'
  END AS age_group,
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL
ORDER BY churn_rate_percentage DESC;

```

Age Group	Churn Rate %
50s	56.04
40s	30.79
60s+	27.95
30s	10.88
Under 30	7.56

5. Churn Rate by Product Holding

```

SELECT
  "num_of_products" AS product_holding,
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL
ORDER BY product_holding ASC;

```

Products Held	Churn Rate %
1	27.71
2	7.58
3	82.71
4	100.00

6. Average Credit Score and Balance (Churned vs Retained)

```
SELECT
  "churned" AS churn_status,
  ROUND(AVG("credit_score"), 2) AS avg_credit_score,
  ROUND(AVG("balance"), 2) AS avg_account_balance
FROM BANK_CHURN_CLEANED
GROUP BY ALL;
```

Churn Status	Avg Credit Score	Avg Balance
Exit	645.35	91,108.54
Retain	651.85	72,745.30

7. Average Tenure and Age (Churned vs Retained)

```
SELECT
  "churned" AS churn_status,
  ROUND(AVG("tenure"), 2) AS avg_tenure,
  ROUND(AVG("age"), 0) AS avg_age
FROM CLEAN.BANK_CHURN_CLEANED
GROUP BY ALL;
```

Churn Status	Avg Tenure	Avg Age
Retain	4.86	37
Exit	4.87	45

Summarize Findings

KPI	Insights
Churn Rate	20.37% of customers churned
Country	Germany has the highest churn (≈32%)
Gender	Slightly higher churn among females (needs significance testing to confirm as a real driver)
Age	Older customers (40+) show more attrition
Activity	Inactive customers churn roughly 2–3× more than active ones
Product Holding	3–4-product customers show the highest churn rates — counterintuitive; worth investigating whether this reflects a small sample size in those groups before treating it as a strong signal

Business Questions (Part 2)

1. Top 10 Customers with Highest Churn Risk

```

/*
credit_score: weight = 0.4
balance:      weight = 0.3
tenure:       weight = 0.2
active_member: weight = 0.1
*/

WITH churn_score AS (
  SELECT
    bc."customer_id", ci."SURNAME", bc."country_name", bc."gender", bc."age",
    bc."credit_score", bc."balance", bc."tenure", bc."active_member",
    ROUND(
      (0.4 * (1 - bc."credit_score" / 1000.0)) +
      (0.3 * (bc."balance" / (SELECT MAX("balance") FROM bank_churn_cleaned))) +
      (0.2 * (bc."tenure" / (SELECT MAX("tenure") FROM bank_churn_cleaned))) +
      (0.1 * (CASE WHEN bc."active_member" = 'Active Member' THEN 0 ELSE 1 END))
    , 2) AS churn_risk_score
  FROM BANK_CHURN_CLEANED bc
  JOIN CUSTOMERINFO ci ON bc."customer_id" = ci."CUSTOMERID"
  WHERE bc."churned" = 'Retain'
),
ranked AS (
  SELECT *, DENSE_RANK() OVER (ORDER BY churn_risk_score DESC) AS customer_risk_rank
  FROM churn_score
)
SELECT * FROM ranked WHERE customer_risk_rank <= 10 ORDER BY customer_risk_rank ASC;

```

Results: see “Top 10 High Risk” CSV stored in GitHub.

2. Churn Trend by Year

```

SELECT
  EXTRACT(YEAR FROM "BANK_DOJ") AS year,
  COUNT(*) AS total_customers,
  SUM("exit_id") AS total_customers_churned,
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM bank_churn_cleaned
GROUP BY ALL
ORDER BY year DESC;

```

Year	Total Customers	Churned	Churn Rate %
2019	3,302	655	19.84
2018	2,591	523	20.19
2017	2,137	478	22.37
2016	1,965	381	19.39
2015	5	0	0.00

3. Rank Countries by Churn Rate and Average Credit Score

```
WITH churn_rate AS (
  SELECT "country_name",
    ROUND(AVG("credit_score"), 2) AS avg_credit_score,
    ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
  FROM bank_churn_cleaned GROUP BY ALL
)
SELECT *, RANK() OVER (ORDER BY churn_rate_percentage DESC) churn_rank
FROM churn_rate;
```

Country	Avg Credit Score	Churn Rate %	Rank
Germany	651.45	32.44	1
Spain	651.33	16.67	2
France	649.67	16.15	3

4. Gender-Based Churn Analysis per Country

```
SELECT "country_name", "gender",
  SUM(CASE WHEN "churned" = 'Exit' THEN 1 ELSE 0 END) AS total_churn,
  ROUND(SUM(CASE WHEN "churned" = 'Exit' THEN 1 ELSE 0 END) * 100 / COUNT(*), 2) AS
churn_rate_percentage
FROM bank_churn_cleaned
GROUP BY ALL;
```

Country	Gender	Total Churn	Churn Rate %
France	Male	350	12.71
France	Female	460	20.34
Spain	Female	231	21.21
Germany	Female	448	37.55
Germany	Male	366	27.81
Spain	Male	182	13.11

5. Loyal Customers (Active, Tenure > 5 Years, No Churn)

```
SELECT ci."SURNAME", bc."country_name", bc."age", bc."balance", bc."estimated_salary"
FROM BANK_CHURN_CLEANED bc
JOIN CUSTOMERINFO ci ON bc."customer_id" = ci.customerid
WHERE bc."tenure" >= 5 AND bc."active_member" = 'Active Member' AND bc."churned" = 'Retain'
ORDER BY bc."country_name", bc."estimated_salary" DESC;
```

Results: see “Loyal Customers” CSV stored in GitHub.

6. Correlation Between Age & Balance (Churned Customers)

```
SELECT ROUND(
  (COUNT(*) * SUM("age" * "balance") - SUM("age") * SUM("balance")) /
  SQRT(
    (COUNT(*) * SUM(POWER("age", 2)) - POWER(SUM("age"), 2)) *
    (COUNT(*) * SUM(POWER("balance", 2)) - POWER(SUM("balance"), 2))
  ), 3) AS correlation_age_balance
FROM BANK_CHURN_CLEANED
WHERE "churned" = 'Exit';
```

correlation_age_balance

-0.023

7. Average Salary & Balance by Product Count and Churn Status

```
SELECT "num_of_products", "churned",
  ROUND(AVG("estimated_salary"), 2) AS avg_salary,
  ROUND(AVG("balance"), 2) AS avg_balance, COUNT(*) AS total_customers
FROM BANK_CHURN_CLEANED
GROUP BY ALL
ORDER BY "num_of_products", "churned";
```

Products	Status	Avg Salary	Avg Balance	Customers
1	Exit	100,639.83	92,028.82	1,409
1	Retain	99,045.36	101,052.82	3,675
2	Exit	100,883.31	90,252.36	348
2	Retain	100,416.63	48,731.13	4,242
3	Exit	106,776.61	85,853.09	220
3	Retain	92,560.21	25,744.26	46
4	Exit	104,763.72	93,733.14	60

Summarize Findings

KPI	Insights
Top 10 High-Risk Customers	Non-active customers with low credit scores (below 560) and high balances (≈180K+) show the highest churn risk, especially in Germany and France.
Churn Trend by Year	Churn rate peaked in 2017 (22.4%) and gradually declined to 19–20% in recent years, suggesting improved retention since 2018.
Country Churn Ranking	Germany records the highest churn rate (32.4%); average credit score is essentially flat across countries, so churn there is more likely driven by non-financial factors such as customer experience or market competition rather than creditworthiness.
Gender-Based Churn by Country	Female customers show higher churn rates across all three countries, most notably in Germany (37.6%) — a pattern worth confirming statistically before building a campaign around it.
Loyal Customers	The majority of loyal, long-tenure, active customers are based in France with high estimated salaries, reflecting strong retention among financially stable, long-term clients.
Age & Balance Correlation (Churned)	Correlation coefficient of -0.023 indicates no meaningful relationship between age and balance among churned customers.
Salary & Balance by Product Count	Customers holding 3–4 products show the highest churn rates despite higher balances — sample sizes for these groups are small (46–220 customers), so this should be treated as a hypothesis to validate with more data, not a confirmed driver.

Insights & Recommendations

Insight	Recommendation
Germany shows the highest churn	Investigate customer satisfaction or competitive offers in Germany.
Older customers (40–50) leaving more	Offer long-term loyalty or tenure-based benefits.
Inactive members more likely to churn	Launch re-engagement campaigns for inactive users.
1-product customers have high churn relative to 2-product customers	Encourage cross-selling (credit card, savings).
High-balance, low-activity, low-credit-score customers show the highest risk	Implement targeted retention campaigns with personalized financial advice or exclusive perks.
Churn peaked in 2017 and improved after 2018	Analyze what post-2018 changes helped, and reinforce them.
Female customers show higher churn, notably in Germany	Investigate via a proper statistical test first; if confirmed, consider engagement initiatives tailored to this segment.
3–4-product customers show the highest churn despite higher balances	Investigate further given small sample size; if confirmed, simplify multi-product management and consider bundled discounts.

Visualization & Power BI Development

1. Snowflake – Power BI Connection

“As a report developer, I want to connect Power BI to Snowflake so that I can visualize live, cleaned data for analysis and reporting.”

Data Source Setup

Parameter	Value
Warehouse	BANK_ANALYTICS_WH
Database	BANK_ANALYTICS_DB
Schema	ANALYTICS
Tables	DIM_ACTIVE_CUSTOMER, DIM_CREDIT_CARD, DIM_CUSTOMER, DIM_EXIT_CUSTOMER, DIM_GENDER, DIM_GEOGRAPHY (Import); FACT_CUSTOMER_CHURN_V1 (Direct Query)
Connection Type	Import (stable refresh) or DirectQuery (live updates)
Authentication	Username & Password or SSO

Direct Query Fact Table (M Language)

```
let
    Source = Snowflake.Databases("<account>.snowflakecomputing.com", "BANK_ANALYTICS_WH"),
    BANK_ANALYTICS_DB = Source{[Name="BANK_ANALYTICS_DB", Kind="Database"]}[Data],
    ANALYTICS_Schema = BANK_ANALYTICS_DB{[Name="ANALYTICS", Kind="Schema"]}[Data],
    FACT_CUSTOMER_CHURN_V1 = ANALYTICS_Schema{[Name="FACT_CUSTOMER_CHURN_V1", Kind="Table"]}[Data]
in
    FACT_CUSTOMER_CHURN_V1
```

Data Validation Checklist

Validation	Expected	Status
Row Count	10,000	Passed
Column Count	13	Passed
Data Type Consistency	Numeric / Date / Binary / Varchar verified	Passed
Null / Blank	No tolerance	Pending — see RAW-layer note earlier in this document
Sample Records	Match between Snowflake and Power BI preview	Passed

2. Data Enrichment

“As a data analyst, I want to create derived columns (like age group, tenure level, and balance segment) so I can analyze customer behavior in meaningful categories.”

Feature	Type	Logic / Formula	Purpose
age_group	Calculated column	Age <30 → Young; 30–40 → Mid-Age; 41–50 → Mature; >50 → Senior	Age-based segmentation
tenure_group	Calculated column	Tenure ≤3 → New; 4–6 → Mid Term; >6 → Long Term	Loyalty segmentation
balance_flag	Calculated column	Bins by percentile (Has Balance / No Balance)	Financial segmentation
score_band	Dimension table	<600 Poor; 600–700 Fair; 701–800 Good; >800 Excellent	Credit segmentation
churn_reason	Dimension table	See disclosure below	Churn driver reporting

3. Analytical Report

“As a report developer, I want to create analytical Power BI reports and DAX-based KPIs so that business users can easily understand customer churn patterns and performance metrics.”

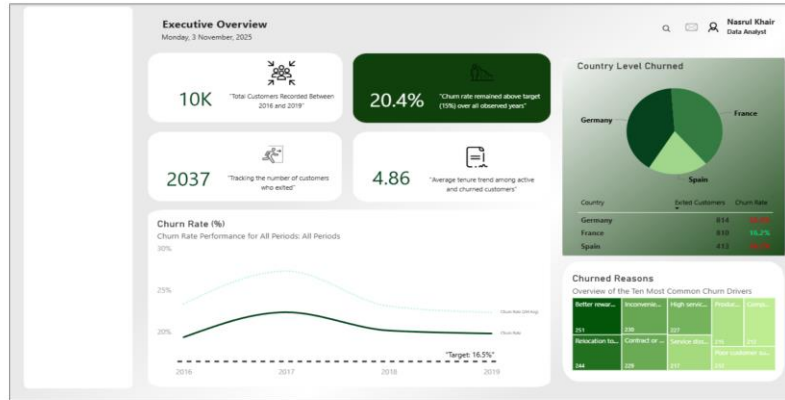
Key DAX Measures

Measure	Formula	Description
Age Group	SWITCH(TRUE(), [Age]<26,"18–25",[Age]<36,"26–35",[Age]<46,"36–45",[Age]<56,"46–55","56+")	Categorize customers into age bins
Churn Count	CALCULATE(COUNTROWS(FACT_CUSTOMER_CHURN_V1), DIM_EXIT_CUSTOMER[ExitCategory]="Exit")	Counts churned customers
Churn Rate by Attribute	DIVIDE([Churn Count],[Total Customers])	Context-specific churn rate
Active %	DIVIDE([Active Members],[Total Customers])	Share of active members
Credit Card %	DIVIDE([Credit Card Holders],[Total Customers])	% of customers with credit cards

Dashboard Design & Layout

1. Executive Overview

Provides a high-level summary of customer churn performance and key retention indicators — the management landing page, with drill-through options to deeper insights.



Key Highlights

- 10,000 total customers were recorded between 2016 and 2019.
- 2,037 customers exited, resulting in a 20.37% churn rate, which remains above the 15% target throughout all observed years.
- The average customer tenure among both active and churned customers is 4.86 years, showing moderate retention.

Country-Level Insights

- Germany recorded the highest churn rate at 32.4%, followed by Spain (16.7%) and France (16.2%).
- No country achieved a churn rate below the overall 16.5% average.

Churn Drivers

Simulated churn reasons include:

- Better rewards elsewhere (251 cases)
- Relocation (244)
- Inconvenient service process (230)
- High service cost (227)
- Service dissatisfaction (217)
- Poor customer support (212)

If this were real data, these categories would suggest pricing, service experience, and support quality as leading churn contributors — which is the hypothesis a future exit-survey dataset should be built to test.

2. Customer Demographic & Profile Analysis

To analyze customer churn patterns through demographic and behavioral segmentation — age, gender, geography, activity status, and credit card ownership — and uncover which customer groups contribute most to churn.

Key Insights

- Age-based churn trends: customers aged 25–45 represent the largest and most churn-prone segment.
- Gender: female customers show a slightly higher churn rate (~21% in this segment); confirm with a significance test before treating as a strategic driver.
- Geography vs. credit profile: average credit score is essentially flat across countries (649.67–651.45), so Germany's higher churn is unlikely to be a credit-score story.
- Activity status: inactive members churn roughly 2–3× more than active ones — the clearest, most consistent signal in this dataset.
- Credit card ownership: non-cardholders churn more, suggesting a cross-sell opportunity.
- Tenure distribution: most customers fall within 3–5 years of tenure, a stage where retention interventions may have the most impact.

Narrative Summary

The demographic analysis reveals that churn is concentrated among mid-age customers (25–45) and inactive, non-credit-card holders. Behavioral engagement and product adoption are the clearest, most defensible levers for retention based on this data; credit score and geography effects are weak and should be investigated further before being treated as drivers. Future initiatives should focus on activating dormant customers, incentivizing credit card usage, and personalizing retention campaigns for high-risk profiles — backed by a proper classification model rather than manually chosen risk weights.